

Mohaddeseh (Sahar) Hosseinzadeh

My work focuses on medical AI, digital health and neuroscience, particularly on connecting computational frameworks with clinical impact. Currently, I am developing ECG models for myocardial infarction detection at the Barcelona Supercomputing Center. I apply deep learning to large-scale neural and cardiac datasets and I have worked with fMRI, EEG, ECG, EMG, and LFP data across the full research pipeline, including literature review, method selection, data auditing and preprocessing, model development and evaluation, and deployment as usable services.

PORTFOLIO : Synapticsahar.github.io [Github](https://github.com) [LinkedIn](https://www.linkedin.com/in/mohaddeseh-hosseinzadeh)

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Research interests: [Medical AI](#) [Digital health](#) [Computational Neuroscience](#) [Computational modeling for clinical applications](#) [Biomedical Signal Processing](#)
[Deep Learning](#)

EDUCATION

 **M.Sc., Computational Neuroscience (Brain and Cognition)** 📅 2025 – July 2026
Pompeu Fabra University, Barcelona (GPA 8.38/10)

 **B.Sc., Industrial Engineering** 📅 2017 – 2023
Sharif University of Technology, Tehran
• Teaching Assistant, Linear Algebra (40+ undergraduates).

RESEARCH EXPERIENCE

 **AI Research Engineer Intern – ECG Foundation Model for STEMI/NSTEMI detection from 12-Lead ECG (RAPID AIM project)**

Barcelona Supercomputing Center, Digital Health Unit (Supervisor: Dr. Paula Petrone) 📅 05/2026 – present

- Tested whether ECGFounder outperforms machine learning models and CNN-based architectures trained from scratch on a private hospital dataset for detecting STEMI and NSTEMI under a shared evaluation protocol on BSC HPC. This distinction matters clinically because STEMI can often be identified from the ECG and trigger urgent reperfusion, whereas NSTEMI diagnosis usually requires additional clinical assessment and cardiac troponin evidence.
- Selected methods through a comprehensive literature review, then audited and preprocessed a private Hospital Sant Pau cohort of 10,491 12-lead ECG recordings.
- Assessed candidate architectures on licensing and commercial-use terms alongside performance. The selected model can therefore be deployed and developed further at the hospital rather than restricted to research use.
- Evaluated performance using AUROC, AUPRC, and sensitivity at fixed specificity, including subgroup analyses, and developed a documented, reproducible pipeline for further clinical development at the hospital.

 **Master's Thesis – Brain States in Vortex Space** 📅 12/2025 – 07/2026
Supervisors: Prof. Gustavo Deco and Prof. Rubén Moreno-Bote, UPF

- Tested whether a richer, more flexible brain-state repertoire in vortex space is associated with better cognition at rest and modelled state transitions using 7T resting-state fMRI from 182 Human Connectome Project participants (900 volumes each; TR 1s; Schaefer parcellation coarsened to 100 cortical regions).
- Built a seven-state synchronization atlas in MATLAB using a local Kuramoto order parameter, 50 principal components (73.9% variance), and k-means over 163,800 pooled volumes; adjusted Rand index reached 0.82 across resampled time points and 0.64 across independent participant halves.
- Across 28 planned tests of seven cognitive outcomes, each using 100,000 participant-label permutations and 10,000 bootstraps, transition and occupancy entropy associations were positive in 26 of 28 tests across general cognitive ability, memory, executive function, and language (strongest: occupancy entropy with the general factor, $r = 0.20$, $p_{\text{perm}} = 0.006$).
- Ridge-dominant elastic net on 49 transition-matrix features reached mean cross-validated r of 0.24–0.29 across five cognitive targets under 100 repeats of 10-fold cross-validation; support-vector regression, random forests, and gradient boosting showed no consistent advantage under 500 label-shuffle permutations with FDR correction.
- Designed a novel exponential distance model that reproduced the off-diagonal group transition matrix at $r = 0.90$ off-diagonal, beat all 5,040 exact state-label permutations, and generalized at $r = 0.54$ under leave-one-participant-out cross-validation, showing preferential transitions to nearby states in vortex space.
- Implemented and evaluated Maximum Occupancy Principle (MOP) models testing whether transitions favour states that preserve access to more future states.

[Thesis PDF](#) | [GitHub Repository](#) | [Presentation Slides](#)

 **Research Collaborator – Sharp Wave Ripple Detection from Hippocampal LFP** 📅 03/2025 – 06/2026
CMC Lab, TU Dresden (Dr. Shervin Safavi)

- Developed an end-to-end feature pipeline for cross-session sharp wave ripple detection. Short-time Fourier transform converted hippocampal LFP into spectrograms, and non-negative matrix factorization extracted recurring spectral components and their time-varying activations.
- Evaluated on the four-session 4-uLED dataset released with rippl-AI, containing four mice, 86.7 minutes of recording and 3,069 expert-annotated ripple events.
- Reproduced the rippl-AI reference models on raw LFP as controlled baselines and benchmarked SVM, dense Informer, CNN1D, InceptionTime, and temporal convolutional networks on factorization-derived features using identical data splits and event-level evaluation criteria; all feature-based models outperformed the strongest reproduced raw-LFP baseline.

- Developed an EEG-based affective-state decoder to read shopper responses to product presentations using a wearable g.tec EEG system paired with eye-tracking glasses. Gaze fixation linked each EEG segment to a product, while pupil diameter provided an arousal measure when self-report ratings were unavailable.
- Built the EEG pipeline end to end using bandpass filtering, ICA-based artifact removal and epoching in MNE-Python, followed by theta, alpha, beta and gamma band-power features.
- Trained EEGNet for four-class valence-arousal classification on DEAP under a leave-one-participant-out protocol. Mean accuracy reached 58.0% against a 35.8% majority-class baseline, with mean macro $F_1 = 0.54$ across 32 folds.
- Characterized the domain shift between laboratory EEG and the wearable deployment setting, including fewer channels, a different montage, ambulatory motion artifacts and pupil-derived arousal
- Deployed the trained model as a real-time inference service: a FastAPI backend, a Streamlit frontend, containerized with Docker on AWS EC2.

- Built the preprocessing stack for EEG and EMG streamed live from sensor-integrated garments, addressing movement artifacts caused by electrode displacement and slow contact-impedance drift that fixed filter settings cannot follow.
- Developed an eye-movement tracking model using headband signals that achieved 91.6% accuracy, and a recurrent neural network that mapped forearm EMG from glove hardware to continuous movement commands for AR/VR control with $R^2 = 0.75$.

- Reproduced [Oroojlooyjadid et al.](#), which applies deep reinforcement learning to the beer game, a four-stage supply chain in which each agent sees only local information, cannot communicate, and shares a single objective with the other three. Standard DQN assumes one fully observed agent playing against an opponent and does not apply.
- Implemented the paper's reward-shaping solution. Once an episode ends, every reward stored during it is adjusted by the difference between the average cost of the whole chain and the agent's own average cost. The agent still acts on local information alone and learns to lower the cost of the chain rather than its own.
- Evaluated the **SRDQN** agent controlling the warehouse against Sterman agents at the other three stages over 50 held-out 100-period games and benchmarked against a base-stock policy.

Code: [Github](#)

SELECTED PROJECTS

Agentic ECG Decision Support

- Tested whether grounding an LLM in a calibrated ECG classifier and guidelines reduces unsupported claims versus retrieval alone and aligns claims with classifier confidence.
- Fine-tuned ECGFounder on PTB-XL (21,799 10-second 12-lead ECGs from 18,869 patients at 500,Hz) to classify normal, myocardial infarction, ST/T change, conduction disturbance and hypertrophy; registered it as an LLM tool.
- Built a system combining classifier output, FHIR conditions, medications and laboratory values, and ESC and ACC/AHA guidelines into schema-validated cited reports with missing-data flags and confidence-based abstention.
- Benchmarked no retrieval, standard RAG, reranked RAG, a tool-using agent and projected ECGFounder embeddings on adversarial PTB-XL and Synthea data. Measured retrieval recall, citation faithfulness, hallucinations, unsupported claims, abstention accuracy, validity, latency and cost, with CI regression gates.

Seizure Detection with a Fine-Tuned Time-Series Foundation Model

- Fine-tuned PatchTST for seizure detection, scored by sensitivity and false alarms per hour rather than accuracy. Seizures occupy well under one percent of a long EEG recording, and a model that never predicts one is still right almost always.
- PatchTST cuts each channel into short patches and processes each channel on its own rather than mixing them at the input. This keeps the model from tying itself to one particular electrode layout.
- Trained and tested on CHB-MIT Scalp EEG from PhysioNet: 22 pediatric patients recorded across 23 cases, roughly 969 hours of continuous recording containing 198 seizures with onset and offset marked by clinicians, sampled at 256 Hz across 23 channels. Benchmarked against clinical baselines on the same data.
- Deployed as a working service: a FastAPI backend, a Streamlit frontend, running on AWS SageMaker.

EEG → Image Decoding via CLIP (THINGS-EEG2)

- Asked whether the category of a picture someone is looking at can be read from their EEG, by placing the brain signal in the same space a vision-language model uses to represent images.
- Trained a Transformer encoder to map each EEG segment onto the CLIP image embedding of the picture that produced it, using a contrastive loss (NT-Xent) that pulls matching pairs together and pushes mismatched ones apart. Retrieved the three most likely image categories for each segment on THINGS-EEG2, and used GPT-4o to turn those categories into a description of what the person appeared to be seeing.

Synaptic Plasticity Modeling in Mouse V1

- Asked whether reward controls learning in visual cortex indirectly, through VIP interneurons inhibiting SST interneurons and releasing the excitatory cells from inhibition.
- Built spiking neural network models in Brian2 using spike-timing-dependent plasticity whose learning rate is scaled by a reward signal.
- Used Allen Brain Observatory two-photon recordings from mouse V1 to compare neural activity during rewarded and passive viewing.

3D U-Net for Brain Tumor Segmentation

- Developed and trained a 3D U-Net for volumetric brain tumor segmentation using MRI scans from the BraTS2020 dataset.
- Used three-dimensional convolutions to learn spatial features across adjacent MRI slices rather than processing each slice independently.
- Optimized voxel-level overlap between predicted and ground-truth tumor regions using Dice loss.
- Applied data augmentation and learning-rate scheduling during training to improve generalization and stabilize optimization.

TRAINING

Computational Neuroscience Summer School

Neuromatch Academy

📅 2024

- Asked whether the variability of the fMRI signal over time carries information about what a person is doing that the raw signal level does not.
- Computed the Shannon entropy of the BOLD time series for each brain region and fed those values to a Gated Recurrent Unit network that predicted which task the participant was performing.
- Tested by transfer across people, training on one group and testing on participants the model had never seen. Entropy features outperformed the raw signal on every run.
- Code: [Github](#) Presentation: [Here](#) Certificate: [Here](#)

ADDITIONAL EXPERIENCE

- **Full-Stack Software Development**, Rahnema College (2021 – 2022). Built and deployed a full-stack web application end to end.

HONOURS AND AWARDS

- Ranked 200th nationally among approximately 150,000 candidates in the Mathematics and Physics group of the 2017 Iranian Nationwide University Entrance Examination (Konkur).

TECHNICAL SKILLS

- **Scientific programming**: Python, MATLAB, C/C++, R, Shell
- **Machine learning**: PyTorch, TensorFlow, scikit-learn, SciPy; CNNs, RNNs and GRUs, Transformers, spiking networks, reinforcement learning, foundation models, PEFT and LoRA
- **Neurophysiology and signals**: EEG, ECG, EMG, LFP, fMRI; MNE-Python, Brian2, Allen SDK
- **LLMs**: HuggingFace, LangChain, RAG, OpenAI API, LLM tracing
- **Computing**: BSC HPC cluster, AWS (EC2, SageMaker, Lambda), Linux
- **Engineering**: Git, Docker, CI/CD, FastAPI, REST APIs

REFERENCES

[Prof. Gustavo Deco](#), MSc thesis supervisor

[Dr. Paula Petrone](#), RAPID-AIM project lead

[Prof. Rubén Moreno-Bote](#), MSc thesis co-supervisor

[Dr. Shervin Safavi](#), supervisor, sharp wave ripple project

[Dr. Milad Alizadeh-Meghbrazi](#), supervisor, wearable BCI research

[CNS Lab, UPF](#)

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